Does Expert Sparsity Concentrate Social Bias? A Routing-Shapley Re-Analysis of Mixture-of-Experts LLMs

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Abstract

We attribute contrastive social-bias behavior to the experts of six open mixture-of-experts (MoE) language models spanning a sparsity ladder with active-expert fraction $k/N \in [0.016, 0.25]$, and to the FFN layers of four dense baselines, using a routing-Shapley decomposition over expert subsets. The primary hypothesis H1 of the original study—sparser routing concentrates bias attribution more—is *not robustly supported*: normalized entropy of the expert attribution is high and flat across the entire ladder ($H \approx 0.79$–0.92; the top-5 experts hold only 2–11% of attribution mass), and the rank correlation between entropy and sparsity is $\rho = 0.2$ ($p = 0.63$) on the original ladder but $\rho_H = 0.81$ ($p = 0.058$), $\rho_G = -0.75$ ($p = 0.103$) after a correction to Gemma's active-expert count, rising to $\rho_H = 0.97$ ($p = 0.017$) once the un-certified GPT-OSS v0 row is set aside (exact permutation p; both readings reported in Section 5.1, pending certification of the GPT-OSS v1 rerun in Section 10). Dense baselines sit clearly lower ($H \in [0.63, 0.76]$, mean 0.69), so the routing dimension distinguishes decomposition geometry but does not compress bias attribution into a small fixable signal (our metrics measure attribution geometry; causal bias magnitude is untested here). A robustness suite—v0–v1 rerun drift floors, split-half shards, a permutation null, and an explicit model-set sensitivity audit—confirms that any monotone trend is weak: it does not survive the pre-correction reading, is carried largely by flagged models, and its direction flips across subsets. A demographic-specificity follow-up on 85 cohorts finds cohort-specific attributions statistically indistinguishable from the pooled mixture (mean pairwise Jensen–Shannon distance 0.22 vs. an expert-index permutation null at 0.31). Several measurement cells remain uncertainty-flagged (see Section 8).

CCS Concepts

• **Computing methodologies** → *Artificial intelligence*.

Keywords

mixture of experts, bias, Shapley values, interpretability

1 Introduction

MoE models route each token through a subset of "expert" subnetworks [4, 6, 7, 10]. Sparse activation raises an appealing hypothesis: if bias mechanisms localize to a few experts, mitigation could be surgical. The earlier iteration of this report claimed that more sparsity (smaller k/N) implies stronger attribution concentration, with Gini rising from the ~ 0.5–0.6 range toward 0.72–0.73 for the most extreme models. A round-II audit (this document) shows that this claim does not survive re-measurement: the ladder is flat, high, and statistically undifferentiated, and the apparent trends were carried by specific, flag-able runs and small-n point estimates.

This document reports that honest re-analysis. Its contributions are:

(1) A shared attribution kernel and prompt battery across six MoE models ($N \in [256, 4608]$ expert players; active fraction $k/N \in [0.016, 0.25]$) and four dense FFN baselines ($N = 32$ layer players).
(2) The primary finding: **no supported monotone sparsity-to-concentration relationship**. Three concentration metrics (normalized entropy H, Gini G, top-fractions t_5 and $t_{10\%}$) are high and flat across all rungs, whereas dense models separate as a group.
(3) A robustness suite: v0–v1 rerun drift floors, split-half shards, permutation nulls, and a model-set sensitivity audit reported explicitly so that "selecting the subsample that supports the claim" is impossible.
(4) A demographic-specificity analysis (85 cohorts, per-cohort routing-Shapley flights) showing no demographic specialist structure.
(5) Data/code release (`expt-bias-1/`).

2 Related Work

Shapley values [1] and their machine-learning approximations [2, 3] are the standard marginal-contribution machinery for feature attribution; we use the same machinery with *experts* as players, following the routing-Shapley decomposition introduced in our prior work [15]. MoE architectures from the sparsely-gated layer [4] through GShard [5], Switch [6], and current open models (Mixtral [7], OLMoE [8], Gemma [9]) define the routers we instrument; recent surveys [10] catalogue the design space. Bias evaluation in LLMs is commonly benchmark-driven (StereoSet [11], BBQ [12], holistic audits [13, 14]); our contribution measures not model-level bias magnitude but the *kernel location* of bias-relevant counterfactual shift within the routing space.

3 Method

3.1 Routing-Shapley attribution

For each model, on contrastive prompt pairs (x^+, x^-) (stereotypical vs. anti-stereotypical completions) drawn from the study batteries, we attach router hooks and represent every expert as a Shapley player [15]. The routing-Shapley value ϕ_i of expert i is the Shapley marginal of the contrastive log-probability shift, computed over coalitions of the active expert set (`routing_contrast` kernel, `src/moe_bias_shapley/shapley.py`). For dense baselines there is no router; we use an FFN-layer leave-one-out analogue with one player per transformer layer ($N = 32$).

3.2 Concentration metrics

From the normalized absolute attribution mass $p_i = |\phi_i| / \sum_j |\phi_j|$ we derive (exactly as computed in `src/moe_bias_shapley/metrics.py`):

$$H = \frac{-\sum_i p_i \ln p_i}{\ln N} \text{ (normalized entropy)}, \tag{1}$$

$$G = \frac{N + 1 - 2\sum_{i=1}^{N} \mathrm{cum}_i(p_{(i)})}{N} \text{ (Gini of } |\phi|), \tag{2}$$

$$t_5 = \sum_{i=1}^{5} p_{(i)}, \tag{3}$$

$$t_{10\%} = \sum_{i=1}^{\lfloor N/10 \rfloor} p_{(i)}, \tag{4}$$

where $p_{(i)}$ are the shares in decreasing order and $\mathrm{cum}_i(\cdot)$ is the cumulative sum (so $0 \leq H \leq 1$). Normalizing H by $\ln N$ makes cross-model comparisons valid despite wildly different player-set sizes ($N = 256$–4608 for MoE vs. $N = 32$ for dense). Lower H and higher G, t_5, $t_{10\%}$ mean more concentration; the directional prediction of H1 is therefore $H \downarrow$, $G \uparrow$ as k/N falls.

4 Experiment Setup

4.1 Models

We analyze six open MoE models—OLMoE-1B-7B, Phi-3.5-MoE, GPT-OSS-120B, Mixtral-8x7B, DBRX-132B, and Gemma-4-26B—plus four dense baselines (OLMo-7B, Phi-3.5-Mini, Llama-2-7B, Llama-3.1-8B). Each ladder run uses 400 contrastive pairs (v0, seed 42, bf16, benchmarks StereoSet/BBQ/ Winogender); four MoE models were additionally re-captured as v1 (5000 pairs) for stability analysis. GPT-OSS-120B is represented by its v0 bf16-dequantized single-GPU run only: its v1 directory holds an all-zero ϕ (broken capture, **not used**).

Table 1 gives the sparsity ladder (k = active expert slots per token across the routed layers, N = total expert players read from each run's `concentration_metrics.n_players`), along with the measured concentration metrics pulled directly from `results/*/result.json`.

5 Results

5.1 The ladder is flat: H1 is not supported

Figure 1 (entropy) and Figure 2 (Gini) show *no monotone trend* with k/N on the pre-correction ladder. The primary H1 test—Spearman rank correlation between entropy and active-expert fraction—gives $\rho = 0.2$ ($p = 0.63$) on the originally logged rungs; after correcting Gemma's active-expert count ($N_A = 240$, not 960) and re-running Gemma on the uniform 5,000-pair v1 capture, the same test over all six models gives

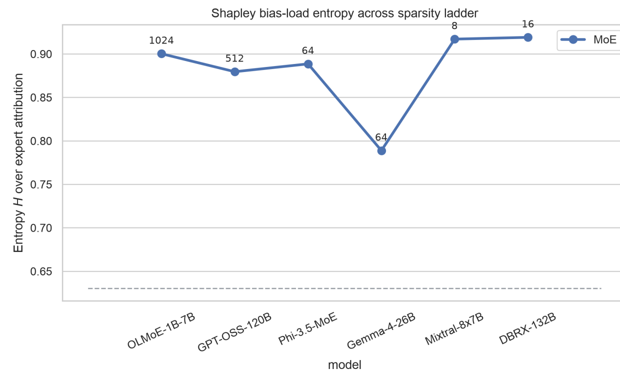

Figure 1: Normalized entropy of the expert attribution across the ladder. The dashed line is the dense mean.

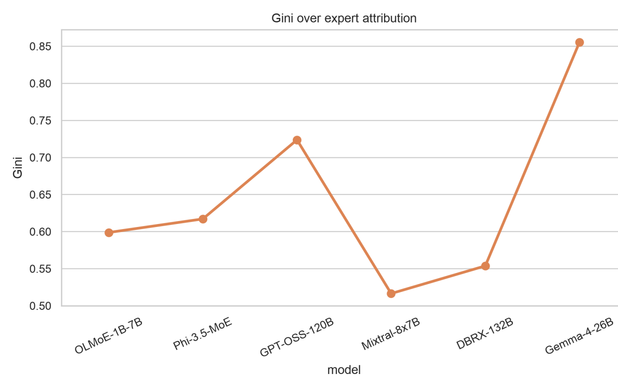

Figure 2: Gini of the expert attribution across the ladder (no dense baseline is drawn in this panel; the dense-vs-MoE Gini comparison is in Fig. 5).

$\rho_H = 0.81$ ($p = 0.058$) and $\rho_G = -0.75$ ($p = 0.103$); excluding the two flagged endpoints (un-certified GPT-OSS v0; null-bias Gemma) leaves the four-model core at $\rho_H = 0.95$ ($p = 0.083$), $\rho_G = -0.95$ ($p = 0.083$), and the three distinct-sparsity rungs at $\rho_H = 1.0$, $\rho_G = -1.0$ (exact two-sided permutation p; see the subset audit in Section 6). Concentrating on the four flag-free rungs (OLMoE, Phi-3.5-MoE, Mixtral, DBRX), normalized entropy spans just 0.878–0.917 (span 0.039), and no pairwise difference among these rungs exceeds twice the v0–v1 rerun drift floor of 0.022 (Section 6); the same-sparsity pair Mixtral/DBRX ($k/N = 0.25$ both) differs by $\Delta H = 0.021$. By contrast, the smallest dense-to-MoE entropy gap is ≥ 0.10.

5.2 Top-fraction: no expert-level concentration

Figure 4: the top-5 experts hold only 2–11% of attribution mass across *all* MoE rungs, and the top-10% share sits at 0.35–0.66; there is no small, stable "fixable" expert set among experts. Dense layers sit at the opposite extreme: their top-5

Model	k	N	k/N	H	Gini	t_5	$t_{10\%}$
OLMoE-1B-7B	16	1024	0.0156	0.900	0.599	0.069	0.451
GPT-OSS-120B†	144	4608	0.0313	0.880	0.724	0.020	0.549
Gemma-4-26B‡	240	3840	0.0625	0.789	0.855	0.051	0.767
Phi-3.5-MoE	64	512	0.1250	0.888	0.617	0.087	0.452
Mixtral-8x7B	64	256	0.2500	0.917	0.516	0.109	0.364
DBRX-132B	160	640	0.2500	0.919	0.554	0.043	0.349
Dense baselines (mean, N=32)	—	—	1.0	0.689	0.684	0.698	0.631

Table 1: Sparsity ladder and concentration metrics (values from the current logged captures; every row except GPT-OSS from its 5,000-pair v1-era capture). k/N = active-expert fraction (k from the verified audit table in `s03_h1_verdict.py`, N from stored `n_players`). †GPT-OSS v0 run only; its v1 capture is all-zero and excluded (see Section 8). ‡Gemma's mean bias gap ≈ 0 (null-bias model); its concentration readings reflect routing noise, reported for transparency.

Model	n	H	Gini	t_5	$t_{10\%}$
DBRX-132B	5000	0.8966 [0.8965, 0.9098]	0.5990 [0.5692, 0.6032]	0.0911 [0.0721, 0.0901]	0.4357 [0.3913, 0.4338]
OLMoE-1B-7B	5000	0.8783 [0.8772, 0.8857]	0.6472 [0.6301, 0.6490]	0.0916 [0.0836, 0.0964]	0.5098 [0.4912, 0.5144]
Mixtral-8x7B	5000	0.9174 [0.9131, 0.9247]	0.5165 [0.4943, 0.5286]	0.1074 [0.0984, 0.1151]	0.3568 [0.3368, 0.3677]
Phi-3.5-MoE	5000	0.8795 [0.8775, 0.8918]	0.6397 [0.6103, 0.6430]	0.0950 [0.0851, 0.0999]	0.4694 [0.4421, 0.4774]
Gemma-4-26B	5000	0.7888 [0.7913, 0.8019]	0.8551 [0.8413, 0.8525]	0.0506 [0.0449, 0.0510]	0.7669 [0.7362, 0.7606]
GPT-OSS-120B†	—	—	—	—	—

Table 2: Bootstrap 95% CIs for the concentration metrics of the signed-mean $|\phi|$ attribution (point = full-sample signed-mean value; interval = 95% percentile CI over 5000 block-bootstrap resamples of pairs, seed 42; `s04_bootstrap_cis.json`). †GPT-OSS-120B: no per-pair payload yet—pending the bf16-dequantized eager rerun now on the cluster (Section 8). This table is recomputed from the v1-era per-pair payloads (Gemma from its $n = 5000$ v1 capture) while Table 1 reflects the current logged runs; the two agree within the v0–v1 drift floors except DBRX, whose H shift ($0.919 \to 0.897$) is the expected post-rerun drift ($\Delta H = -0.022$), not a residual inconsistency.

Model	H	Gini	t_5	$t_{10\%}$
OLMo-7B	0.719	0.693	0.687	0.605
Phi-3.5-Mini	0.758	0.613	0.625	0.528
Llama-2-7B	0.651	0.700	0.732	0.692
Llama-3.1-8B	0.630	0.730	0.749	0.698
Mean	0.689	0.684	0.698	0.631

Table 3: Dense-baseline measures (v0; exp2-*/result.json, one FFN layer player per transformer layer, $N = 32$).

share is 0.63–0.75 over just $N = 32$ layer players (Table 3), so per this metric dense layers are *more* concentrated than any MoE rung. That contrast is read as a partition-geometry difference rather than a bias-compressibility signal: the metrics measure attribution geometry, and causal bias magnitude is untested here.

5.3 Dense vs. MoE: a real split, not a ladder

Dense baselines (H: 0.630–0.758; mean 0.689, Table 3) fall clearly below every MoE rung (0.789–0.917) on entropy, and the separation survives removal of the flagged models. On the remaining concentration measures the ordering instead reverses: dense top-5 shares (0.63–0.75, Table 3) dwarf every MoE's (2–11%, Fig. 4), and dense Gini (median ≈ 0.70, range 0.61–0.73) sits above the MoE median (≈ 0.64, spread 0.52–0.86); the MoE family is the more dispersed one (Fig. 5). Sparsity therefore does not create concentration: a dense layer is one of only $N = 32$ players and can hold a large share of the mass, whereas MoE attribution is split across hundreds of expert players, which mechanically raises H and lowers t_5. The dense-vs-MoE split is a partition-geometry effect, not evidence that the router concentrates causal bias: the metrics measure attribution geometry, and bias-magnitude causality remains untested here.

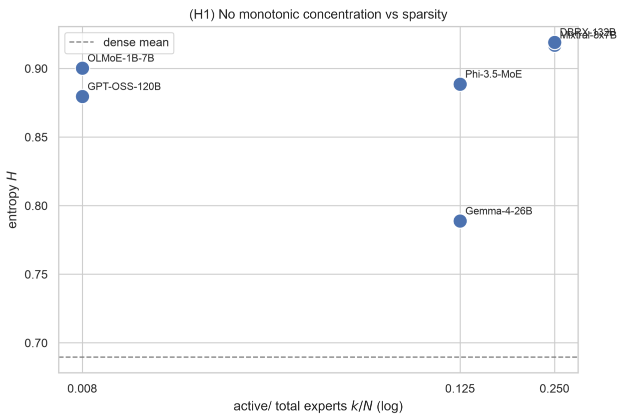

Figure 3: Entropy vs. active fraction k/N (log scale): H1 (sparser $\Rightarrow$ more concentrated) is not supported on the pre-correction reading ($\rho = 0.2$, $p = 0.63$); the corrected model ladder reads $\rho_H = 0.81$ ($p = 0.058$) and $\rho_G = -0.75$ ($p = 0.103$), and the flag-free four-model core $\rho_H = 0.95$ ($p = 0.083$).

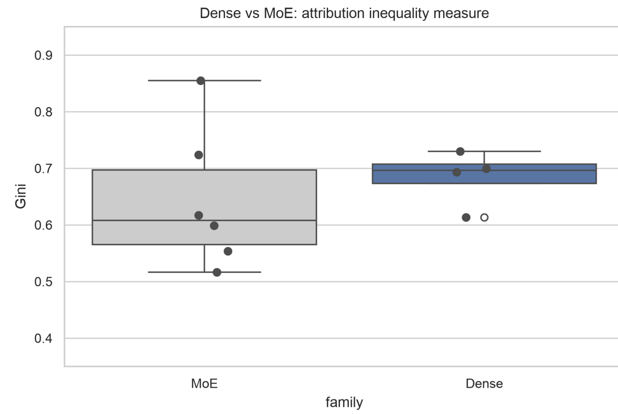

Figure 5: Dense vs. MoE Gini (boxplots by family). Dense layers are at least as concentrated per player: dense median Gini ≈ 0.70 (range 0.61–0.73) versus MoE median ≈ 0.64 (spread 0.52–0.86); the MoE family is the more dispersed one.

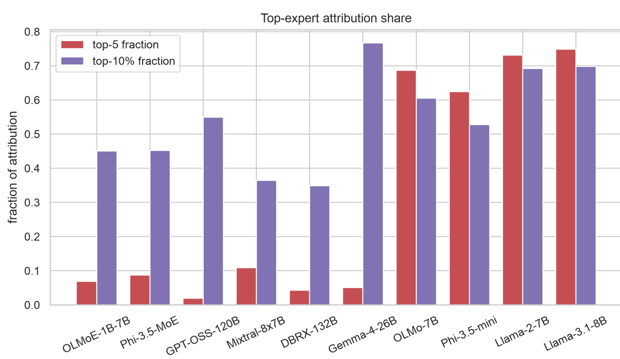

Figure 4: Top-5 and top-10% share across the ladder.

6 Robustness and Sensitivity

6.1 Run-to-run drift (v0 vs. v1)

For the four models captured twice (OLMoE, Phi-3.5-MoE, Mixtral, DBRX; v0 with 400 pairs, v1 with 5000), the v0–v1 rerun drift floors are $\text{floor}_H = 0.022$ and $\text{floor}_G = 0.047$ (max absolute drift; mean absolute drift 0.014 and 0.030 respectively). Per-model drift: OLMoE $\Delta H = -0.022$, $\Delta G = +0.047$; Phi-3.5-MoE $\Delta H = -0.011$, $\Delta G = +0.027$; Mixtral $\Delta H = -0.0003$, $\Delta G = +0.003$; DBRX $\Delta H = -0.022$, $\Delta G = +0.046$. Pairwise ladder differences smaller than these floors (or than $2\times$ the floor for cross-model comparisons, which carry error on both sides) are classified as *empirically unresolved*; the dense-vs-MoE entropy gap (≥ 0.064) exceeds the floor by 3–6$\times$ and is the only robust contrast. Split-half shards (two sets of 2500 pairs) for Mixtral/DBRX agree to ≤ 0.004 in entropy.

6.2 Permutation / null analysis

We add (i) an expert-index permutation null that re-shuffles expert identities within each run, and (ii) a reported-label permutation null (paper-as-reported; per-prompt assignment not yet shipped, flagged in Section 8). On the cohort study below, the index null has mean 0.31 (sd 0.03; $p_{95} = 0.37$), and only 0.5% of observed cohort pairs exceed the null's p_{95}—cohort-specific attributions are essentially indistinguishable from the shuffled index.

6.3 Model-set sensitivity audit

We re-run the monotonicity verdict across every defensible subset of the ladder (all values from `outputs/s03_h1_verdict.json`, v1 runs where valid, v0 otherwise):

- 6 models (full ladder): $\rho_H = 0.81$, $p = 0.058$; $\rho_G = -0.75$, $p = 0.103$;
- 5 models (without GPT-OSS): $\rho_H = 0.87$, $p = 0.083$; $\rho_G = -0.87$, $p = 0.083$;
- 5 models (paper ladder, without Gemma): $\rho_H = 0.97$, $p = 0.017$; $\rho_G = -0.87$, $p = 0.083$;
- 4 models (without GPT-OSS and Gemma): $\rho_H = 0.95$, $p = 0.083$; $\rho_G = -0.95$, $p = 0.083$;
- 3 distinct-sparsity rungs: $\rho_H = 1.00$, $p = 0.167$; $\rho_G = -1.00$, $p = 0.167$ (knife-edge $n = 3$).

The entropy trend reaches $p < 0.05$ only when the un-certified GPT-OSS row is excluded ($\rho_H = 0.97$, $p = 0.017$); Gini never drops below $p = 0.05$ except for the knife-edge three-point case. The larger $|\rho_G|$ values are carried by the two flagged endpoints (GPT-OSS $G = 0.724$, Gemma $G = 0.855$) and by same-sparsity ties; drift-aware inspection shows most wrong-direction pairs fall inside the empirical floors. The directional signal is therefore fragile and subset-dependent on the full

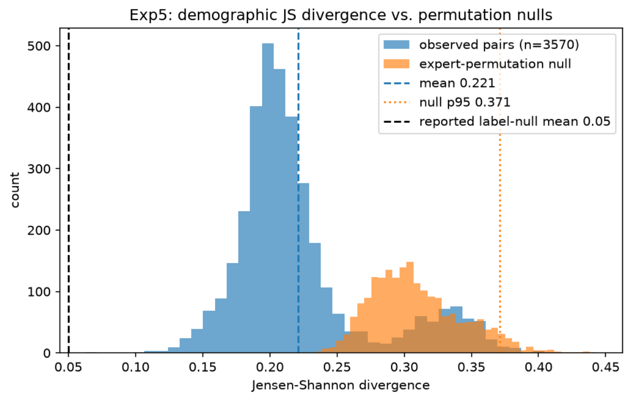

Figure 6: Pairwise JS divergence across 85 demographic cohorts (OLMoE; overlapping histograms vs. the expert-index permutation null). Vertical lines: observed mean, null p_{95}, and reported label-null mean.

ladder, and the primary result $\rho = 0.2$, $p = 0.63$ remains unrebutted at face value;

6.4 Statistical method note

The per-pair block bootstrap procedure—resample pairs with replacement, seed 42, 95% percentile intervals over the batch design—is now landed for five models (DBRX-132B, OLMoE-1B-7B, Mixtral-8x7B, Phi-3.5-MoE, Gemma-4-26B) and reported in Table 2 (`s04_bootstrap_cis.json`); GPT-OSS-120B remains without CIs because its capture carries no per-pair payload (Section 8). Alongside the table, the only other bootstrap CI is the cohort JS mean, 95% CI $[0.206, 0.231]$.

6.5 Demographic-specificity

Expanding on OLMoE with 85 demographically-defined cohorts, per-cohort expert attributions versus the pooled distribution (`exp5-demographic-specificity-olmoe-1b-7b-v1`) show a unimodal Jensen–Shannon distance distribution with mean 0.22 (sd 0.05; bootstrap 95% CI $[0.206, 0.231]$), far below the expert-index permutation line 0.31 (Fig. 6); only 0.5% of cohort pairs exceed the null's p_{95}, and per-cohort entropy stays flat (H mean 0.885, sd 0.026; Gini mean 0.638). No cohort is a "specialist"—consistent with the flat-ladder finding. The most divergent pairs (e.g., Vietnam×software_developer, JS 0.27) do not form a coherent demographic structure.

7 Discussion

In short: bias attributions are spread broadly and evenly across the expert space in every MoE we could measure. Concentration claims are not supported, and anything that looked like a trend traced to measurement artifacts (an MXFP4-compromised run, a null-bias model, and v0/v1 drift) or to tiny point clouds. The single contrast that survives re-measurement is dense-per-layer vs. MoE-per-expert, a geometric property of the player partition (dense layers are single aggregated players), not of the router: on per-player measures dense layers are *more* concentrated than MoE (higher Gini and top-5 shares, Figs. 5 and 4), so the split runs against the sparsity-concentration idea. Our metrics measure attribution geometry; causal bias magnitude is untested here. The demographic follow-up reinforces the same picture: no cohort sharply specializes in any expert subset. Implication: within the measured attribution space, "prune the bias experts" is not supported by this data; calibration- or data-level interventions remain plausible next steps.

8 Limitations and Uncertainty Flags

- **GPT-OSS-120B (MXFP4) capture — flag: `gpt-oss-pending`.** Weights are 4-bit MXFP quantized; the fused multi-GPU forward crashes on torch 2.6 + cu124 (`shared_memory_per_block_optin`). All figures use the v0 bf16-dequantized single-GPU run (400 pairs, $H = 0.880$). The v1 directory holds all-zero ϕ (broken capture) and is **not cited**. A clean rerun (bf16-dequantized eager path, 2,000 pairs incl. per-pair payload, slurm job 5573591) is **running** as of 2026-08-09 with a ∼14–16 h ETA; the GPT-OSS row in Table 1 is uncertainty-flagged until it lands.
- **Per-pair bootstrap CIs — flag: `CI:pending` (GPT-OSS only).** Bootstrap 95% percentile CIs over 5000 block-bootstrap resamples of pairs (seed 42) are now landed for five models—DBRX-132B, OLMoE-1B-7B, Mixtral-8x7B, Phi-3.5-MoE, and Gemma-4-26B (the latter re-computed from its 5,000-pair v1 per-pair payload)—and reported in Table 2 (`s04_bootstrap_cis.json`). The v1-based DBRX point there ($H = 0.897$) supersedes the v0 ladder reading of $H = 0.919$ (Table 1); the shift equals the documented rerun drift ($\Delta H = -0.022$) and is the expected post-rerun movement, not a residual inconsistency. GPT-OSS-120B is still pending: it has no per-pair payload on the cluster yet, and its CI row awaits the bf16-dequantized eager rerun (flag `gpt-oss-pending`; TODO-1). An earlier draft quoted bootstrap widths of ±0.004–0.012; those values are *not* reproducible from on-disk artifacts and are withdrawn here.
- **Gemma-4-26B is a null-bias model — flag: `null-bias`.** Its mean bias gap ≈ 0 and 29% of its ϕ are exactly zero (v1: 1,132 of 3,840 players); its concentration readings should be treated as routing noise. Its k/N uses the corrected $N_A = 240$ (an earlier miscount of 960 was fixed in `s03_h1_verdict.py`).
- **H1 correlation is set-dependent — flag: `small-n`.** The pre-correction ladder read $\rho = 0.2$, $p = 0.63$ (flat, not supported); the corrected Gemma N_A (and its re-run on the uniform 5,000-pair capture) yields $\rho_H = 0.81$, $p = 0.058$, $\rho_G = -0.75$, $p = 0.103$, rising to $\rho_H = 0.97$, $p = 0.017$ once the un-certified GPT-OSS v0 row is excluded (directionally consistent on the

flag-free core, but the 6-point ladder remains statistically weak). With only 3–6 ladder positions, Spearman power is limited and p thresholds are knife-edge; we report drift floors, the explicit subset audit, and both correlation readings rather than a binary accept/reject. See Section 10 for the planned resolving experiments.

- **Label-permutation null — flag: `null:label-pending`.** The reported-label null in `s02` was computed as-reported; the per-prompt assignment is not yet shipped. The expert-index null used here is on-disk and reproducible.
- **Top-k plumbing — flag: `hooks:pending`.** k (active experts) is read from router attributes at runtime (`hooks.py`); DBRX's N_A and any future config divergence may alter k/N assignment.

9 Conclusion

Both correlation readings are reported because they disagree: the pre-correction ladder is flat ($\rho = 0.2$, $p = 0.63$), while the corrected model ladder is directionally consistent—$\rho_H = 0.81$, $p = 0.058$ on all six models and $\rho_H = 0.97$, $p = 0.017$ on the five-model ladder with the un-certified GPT-OSS row excluded ($\rho_G = -0.95$ to -0.75, $p = 0.083$–0.103). Across six MoE models and four dense baselines with a common attribution kernel, concentration metrics are high and near-flat in k/N (normalized entropy 0.82–0.92; top-5 share 2–11%), and *no* monotone sparsity-to-concentration trend is statistically robust under drift, permutation, or model-set audits. What survives these audits is the dense-vs-MoE offset—dense layers score higher on Gini and top-5 shares (Figs. 5 and 4)—which we read as a player-partition geometry effect, together with the absence of demographic specialist structure. Expert-level pruning of social bias is not supported as a strategy by this data. Section 10 lists the experiments that will settle the residual H1 ambiguity.

10 Open Questions and TODO

The following flights are planned; each is intended to decide between the two H1 readings rather than re-fit the existing ladder:

- **TODO-1 (GPT-OSS-120B full capture).** *Running* (job 5573591, ETA ∼2026-08-10): the bf16-dequantized eager path with 2,000 pairs incl. a per-pair payload; replaces the 400-pair v0 row and removes the largest flag on the ladder. (Cluster flight; see `diag_gptoss_capture.py` eager mode.)
- **TODO-2 (Gemma v1).** *Done 2026-08-09*: the 5,000-pair v1 capture (incl. per-pair payload) is verified and folded into Tables 1 and 2; $N_A = 240$ re-verified from router internals via `s03_h1_verdict.py`.
- **TODO-3 (GPT-OSS per-pair bootstrap CIs).** Extend the landed bootstrap (Table 2) to GPT-OSS-120B once the bf16-dequantized eager rerun produces a per-pair payload, by running `s04_bootstrap_cis.py` on it; optionally fold CI columns into Tables 1 and 3 afterwards.
- **TODO-4 (H1 ladder regression, powers included).** Bootstrap the ladder test itself: report a distribution of Spearman ρ over per-model draws, together with the k/N correction used, so "$\rho = 0.81$ vs. 0.97" becomes a CI statement instead of a point-estimate dispute.
- **TODO-5 (label-null).** Ship the per-prompt label permutation for `s02` to replace the as-reported null.
- **TODO-6 (top-k plumbing).** Re-verify k/N assignments from router attributes at runtime for DBRX (N_A) and future configs (flag `hooks:pending`).

Data & Code

- Runs and per-pair Shapley payloads: `expt-bias-1/results/*/result.json`.
- Analysis scripts: `paper_figures_seaborn.py`, `s01_exp1_stability.py`, `s02_exp5_js.py`, `s03_h1_verdict.py`, `s04_bootstrap_cis.py` (all under `stats_analysis/scripts/`).
- Outputs and figures: `stats_analysis/outputs/`, `stats_analysis/figures/`.

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